

Code-Layout, Readability and Reusability

Date	5 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation used.
2	Proper File Structure	Files and folders are logically Organized	Yes	Files organized into folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear and descriptive names.
4	Function & Method Names	Functions are descriptively named	Yes	Functions named by purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Important sections commented.
6	Modular Design	Code is split into reusable functions/module	Yes	Code divided into modules.

7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exceptions handled.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Login Module	Python, Flask	User Authentication	High
2	Prediction Module	Python, Scikit-learn	Crop Prediction	High
3	Database Module	MySQL	Data Storage	High
4	Report Module	Python	Report Generation	Medium
5	User Interface	HTML, CSS	Dashboard	Medium

Reusable Components / Modules:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized
Readability	5	Easy to understand
Reusability	5	Modules can be reused
Documentation / Comments	4	Basic comments included
Overall Score	5	Clean, maintainable code